Week of 27th Nov.

Tuesday, 14 November 2023 10:35 AM

Shift in perspective : Flow profiles (microscopic look) to Large scale Info (Practical)

Navier stoke → $v(r,\theta,z)$

- → Total force on wall
- → Overall flow rate
- → Total work performed by a pump/turbine mixer

Macroscopic Mass Balance

Mass balance on a CV : $\dfrac{dm_{cv}}{dt} + \iint_{CS} \rho(\hat{n}\cdot\underline{v})\,dS = 0$ → CS_1 , CS_2

- → Multiple inlet & outlet
- → Constant density

$\sum \rho_i \iint_{A_i} (\hat{n}\cdot\underline{v})\,dS$

$(\hat{n}\cdot\underline{v}) = \|\hat{n}\|\,\|\underline{v}\|\cos\theta$

1

v

$\underline{v} = v\,\hat{e}_v$

mag.

"velocity" direction (or unit vector)

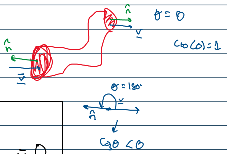
[constant @ each inlet/outlet]

$\sum \rho_i \iint_{A_i} v\cos\theta\,dS = \sum \rho_i \cos\theta \iint_{A_i} v\,dS$

$\boxed{\langle v\rangle} = \dfrac{\iint_{A_i} v\cdot dS}{\iint_{A_i} dS} = \dfrac{1}{A_i}\iint_{A_i} v\cdot dS$

$\therefore \iint_{A_i} v\cdot ds = \langle v\rangle A_i$

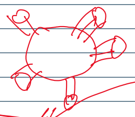
Macroscopic Mass Balance



$$\boxed{\dfrac{dm_{cv}}{dt} + \sum_{i=1}^{\#\,\text{streams}} \left[\rho A\cos\theta\,\langle v\rangle\right] = 0}$$

θ_i = Angle b/w $\hat{n}_i$ & $\hat{v}_i$ (will carry the negative sign depending on inlet & outlet)

$N_o\ \rho$



Macroscopic Momentum Balance

$\dfrac{dP}{dt} + \iint_{CS} \underbrace{(\hat{n}\cdot\underline{v})\,\rho\,\underline{v}\,dS}_{\text{Scalar}} = \sum_{\text{on CV}} f$; $\beta = \dfrac{\langle v\rangle^2}{\langle v^2\rangle}$

$\beta = 1$ (turbulent) ; $\beta = 0.75$ (laminar)

$\sum \rho_i \cos\theta\,\hat{v} \iint v^2\cdot dS = \sum_{i=1}^{\#\,\text{streams}} \rho A\cos\theta\,\langle v^2\rangle\,\hat{v}$

somewhat hard to compute

$\hat{e}_v$

$\langle v\rangle^2 \neq \langle v^2\rangle$

$\langle v\rangle^2 = \left(\dfrac{\iint v\cdot ds}{\iint ds}\right)^2$ $\langle v^2\rangle = \dfrac{\iint v^2\,ds}{\iint ds}$

$\langle v^2\rangle \rightarrow \langle v\rangle^2$

$$\boxed{\dfrac{dP}{dt} + \sum_{i=1}^{\#} \left[\dfrac{\rho A\cos\theta\,\langle v\rangle^2}{\beta}\,\hat{v}\right] = \sum_{\text{on CV}} f}$$

unit vector!

$= f_g + f_{\text{inlet}} + f_{\text{outlet}} + f_{\text{walls}}$

mg

$\iint [\hat{n}\cdot\Pi]\,ds = \iint \hat{n}\cdot[-pI + \underline{\underline{\tau}}]\,dS$

$= \underbrace{\iint -p\hat{n}\,dS}_{\text{pressure dominates!}} + \iint \hat{n}\cdot\underline{\underline{\tau}}\,ds$ 0

(Viscous terms accounted for in f_{wall} !)

f = Force = (Pressure)·A

$= -P_1 A_1$

$f_o = P_2 A_2$

Macroscopic Energy Balance

$\dfrac{dE}{dt} + \iint_{CS} (\hat{n}\cdot\underline{v})\,\rho\,\hat{E}\,dS = Q_{in} - W_{by}$

$\alpha = \dfrac{\langle v\rangle^3}{\langle v^3\rangle}$; $\alpha \approx 1$ (turbulent)

$\alpha = 0.98$

- → Open sys,
- → $E = E_K + E_p + U$
- → constant velocity across inlet/outlet
- → $\hat{H} = \hat{U} + P\hat{V}$
- → Flow work / shaft work
- → No phase change
- → SS, single input/single output
- → $\Delta T \sim 0$, no reaction

$$\boxed{\dfrac{\Delta p}{g\rho} + \dfrac{\Delta\langle v\rangle^2}{g\,2\alpha} + \Delta z + \dfrac{F}{g} = \dfrac{-W_{s,by}}{mg}}$$

(+) Net head

$= \widehat{H}_d$

$= H_d$

$+ h_f$

$+ h_p$

$+ h_v$

$+ z$ [=] length

MEB → Mechanical Energy Balance

h + z

$4f\,\dfrac{L}{D}\,\Delta p$

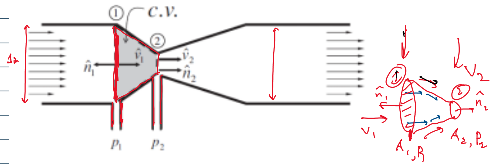
→ closed form

$f = \dfrac{a}{\sqrt{f}}$

$\alpha = \beta = 1$

0 gpm 200 gpm

PROBLEM: 1.21 A Venturi meter with a 1.00 *in* diameter throat is to be installed into a 2 *in* line (Schedule 40 piping, smooth) with water flowing at 25°C. If the flow is turbulent and the range of expected flow rates is 0 – 200 *gpm*, what is the expected range of pressure drop in the Venturi meter? You may neglect friction.



In a Venturi meter, flow is directed through a gently tapering tube. Pressure is measured before the contraction (1) and at the point of smallest diameter (2) (i.e., the throat). The relationship between the measured pressures and the fluid velocity may be deduced from the mechanical energy balance with assistance of the macroscopic mass balance.

2" Schedule 40 pipe: ID = 2.067"

Pg 772 [Ex. 9.5]

① Velocity relationships (v_1 vs. v_2)?

$$\frac{dm}{dt} + \sum_i \rho \cos\theta_i \langle v \rangle_i = 0$$

Mass Balance:

$$\cancel{\frac{dm}{dt}} + \rho_1 A_1 \cos\theta_1 \langle v_1 \rangle + \rho_2 A_2 \cos\theta_2 \langle v_2 \rangle = 0$$

0

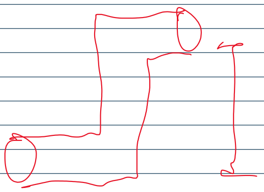
$$\Rightarrow \rho_1 A_1 (-1) \langle v_1 \rangle + \rho_2 A_2 (1) \langle v_2 \rangle = 0$$

$$\rho_1 A_1 \langle v_1 \rangle = \rho_2 A_2 \langle v_2 \rangle \quad ✓$$

$$\boxed{Q_1 = Q_2} \quad (\rho_1 = \rho_2)$$

$$\frac{\pi D_1^2}{4} \langle v_1 \rangle = \frac{\pi D_2^2}{4} \langle v_2 \rangle$$

$$\langle v_1 \rangle = \left(\frac{D_2}{D_1}\right)^2 \langle v_2 \rangle$$



② Pressure drop? (MEB)

$$\frac{\Delta p}{\rho} + \frac{\Delta \langle v \rangle^2}{2\alpha} + \cancel{g \Delta z}^{\,0} + \cancel{F}^{\,0} = \cancel{\frac{-W_{s,by}}{m}}$$

$$\alpha = 1; \quad \frac{\Delta p}{\rho} + \frac{\Delta \langle v \rangle^2}{2} = 0$$

(Out-in)

$$\frac{P_2 - P_1}{\rho} = -\frac{(\langle v_2 \rangle^2 - \langle v_1 \rangle^2)}{2} ; \quad \langle v_1 \rangle = \left(\frac{D_2}{D_1}\right)^2 \langle v_2 \rangle$$

$$\frac{P_1 - P_2}{\rho} = \frac{\rho}{2} \left(\langle v_2 \rangle^2 - \left[\left(\frac{D_2}{D_1}\right)^2 \langle v_2 \rangle\right]^2 \right)$$

$$\langle v_2 \rangle \approx v_2$$

$$\boxed{-\Delta P = \frac{\rho \langle v_2 \rangle^2}{2} \left(1 - \left(\frac{D_2}{D_1}\right)^4\right)}$$

$$v_2 = \frac{Q}{A} =$$

$$\rightarrow \left[\text{Pressure drop} = -\Delta P = -(P_2 - P_1) \right]$$

"neg."

PROBLEM: 1.23 A gasoline tank is connected to a 25 *ft* hose (ID = 1.50 *cm*) as shown in Figure 1.49. The ambient temperature is 38°*C*. What is the maximum height of the barrier over which the gasoline may be siphoned? You may neglect frictional losses. Note the following physical property data: density of gasoline=5.6 lb_m/gal, vapor pressure at 38°*C* is 12.3 *psia*.

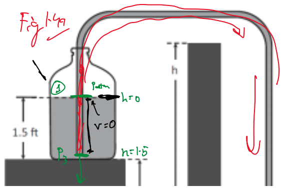
Siphon flow

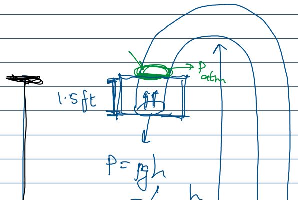
You treat the fluid as a continuum (no breaks)

(free to choose the pressure that is the most convinient)

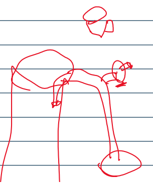
As liquid level is falling (the pressure @ bottom will be changing)

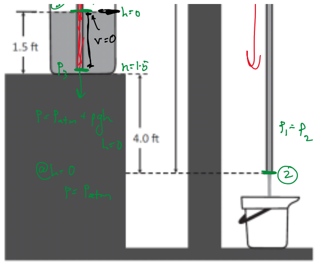
↳ hard to use that

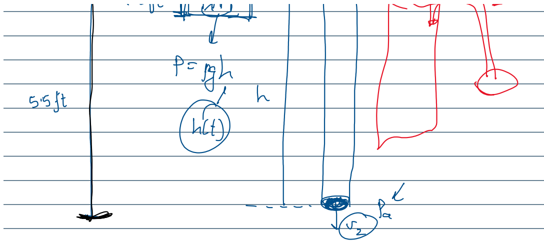




Siphon







↳ hard to use that

① What is the exit velocity?

Mechanical energy balance.

No shaft

$$\frac{\Delta P}{\rho} + \frac{\Delta (v)^2}{2\alpha} + g\,\Delta z + \cancel{F} = -\frac{\cancel{W_{s,by}}}{m}$$

No friction

α = 1 (turbulent)

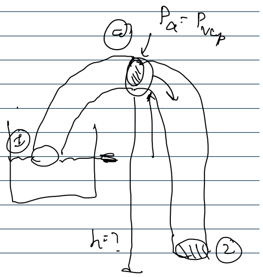
$$\frac{\overset{0}{\cancel{(P_2 - P_1)}}}{\rho} + \frac{v_2^2 - \overset{0}{\cancel{v_1^2}}}{2} + g(z_2 - z_1) = 0$$

$$\frac{v_2^2}{2} = -g(z_2 - z_1)$$

$$\boxed{v_2 = \sqrt{2g\,\underbrace{(z_1 - z_2)}_{5.5}}}$$

② Max height: fluid will stop flowing when suction breaks (i.e. gas/air is introduced somehow) ✓

↳ This can happen when the pressure equals vapor pressure (to "boil" the liquid)



Find the height @ $P = P_{vap}$

$$\frac{\Delta P}{\rho} + \frac{\Delta v^2}{2\alpha} + g\,\Delta z + \cancel{F} = -\frac{\cancel{W_s}}{m}$$

$z_2 - z_1$

$P_{vap} = 12.3$

$$\frac{P_2 - P_a}{\rho} + \frac{\overset{0}{\cancel{v_2^2 - v_a^2}}}{2\alpha} + g(+h) = 0$$

atm = 14.69

$$\boxed{h} = 0$$

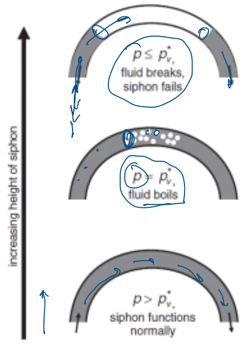
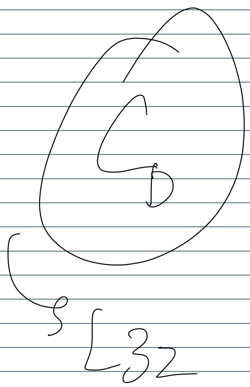
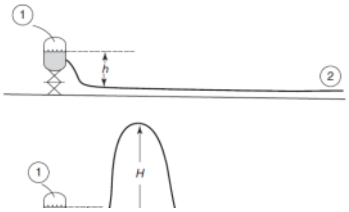
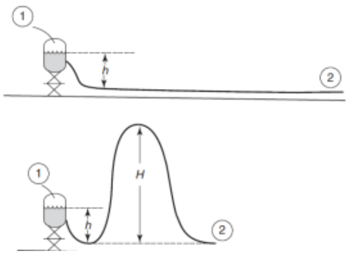

increasing height of siphon
$p \le p_v^*$, fluid breaks, siphon fails
$p = p_v^*$, fluid boils
$p > p_v^*$, siphon functions normally

Figure 1.16 Schematic of the formation of a vapor gap in a siphon. This occurs when the barrier over which the fluid travels rises above a critical height; the critical height may be calculated as shown in the example.



PROBLEM: 1.24 A water tank is connected to a 100 ft hose (ID= 1.50 cm) as shown in the top of Figure 1.50. The height h is 1.8 m. Calculate the average velocity of water in the hose. Do not neglect friction; you may assume turbulent flow.





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